

Biophysics Module

BIOPHYSICS OF SOLUTIONS

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A- STATES OF MATTER

The most classic states are { *liquid state*
solid state
gaseous state

We also include, among others, **intermediate** states such as **liquid crystals** (used in mobile telephony), which combine the properties of a conventional liquid with those of a crystalline solid.

A **plasma**, called 4th state of matter, is an **ionized gas** made up of electrons, ions, atoms, or neutral molecules.

1- THE LIQUID STATE

- ↗ has its own volume,
- ↗ takes the shape of the container that holds it,
- ↗ at rest, its free surface is always flat and horizontal,
- ↗ **condensed** (particles close together) but **fluid** (particles weakly bound, mobile relative to each other).



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2- THE GAZEOUS STATE

- ↷ It has neither a fixed shape nor a fixed volume,
- ↷ tends to occupy all available volume,
- ↷ dispersed, completely disorganized, its particles almost independent,
- ↷ is made up only of atoms and molecules.

B- ENERGY

A property of a system capable of performing work, it is expressed in

- ↷ Joules (J) in the *SI* system (*M.K.S.A*)
- ↷ Ergs (*erg*) in the *CGS* system: $1\text{erg} = 10^{-7}J$
- ↷ Calories (*cal*) : $1\text{cal} = 4.184J$
- ↷ Electrons-volts (*eV*) : $1\text{eV} = 1.6 \times 10^{-19}J$

C- SATURATING VAPOR PRESSURE

1- DEFINITIONS

The system consists of two immiscible liquids.

We have two different **phases** and only one liquid **state**.



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Phase: region of space where we have the **same parameters** and the **same properties** (chemical composition, physical properties, state of matter).

We define,

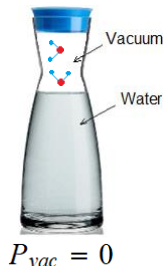
- ↷ **Open system** : it exchanges **energy** and **matter** with the surrounding environment.
- ↷ **Closed system** : it only exchanges **energy** with the surrounding environment.
- ↷ **Isolated system** : **no exchange** with the surrounding environment.



2- SATURATING VAPOR PRESSURE

The closed chamber is at a fixed temperature.

A certain number of water molecules will pass into the state of **vapor** until the vacuum space is saturated and thus reach a pressure called **saturation pressure** (P_{sat}) or **saturating vapor pressure**, which depends only on the temperature.



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P_{sat} : denotes the **vapor** pressure of a substance that is in **equilibrium** with its **liquid phase** at a given temperature.

Equilibrium means that once saturation is reached, the flow of molecules passing from the liquid phase to the gaseous phase is **equal** to that returning from the gaseous phase to the liquid phase.

↪ For a **volatile** (evaporable) substance, we have: $P_{sat} > P_{atm}$

↪ For water, $P_{sat} = \left(\frac{t}{100}\right)^4$; $100^\circ C \leq t \leq 200^\circ C$: **Duperray's formula**

Recalls

$$[P_{sat}] = bar \quad [t] = ^\circ C$$

* $1bar = 10^5 Pa \approx 1atm$

* $1atm = 101325Pa$

* **Pressure exerted by a column of liquid:**

$$P = \frac{\text{Force}}{\text{Surface}} = \frac{mg}{S} = \frac{\rho Vg}{S} = \frac{\rho Shg}{S} \Rightarrow P = \rho gh$$

For $h = 760mm$ of mercury ($\rho = 13.6g/cm^3$), $P \approx 101325Pa$

We write : $760mmHg \approx 101325Pa = 1atm$

* $1Pa = 7.5 \times 10^{-3} mmHg$



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Exercise 1:

One liter of pure water is placed in a pressure cooker. The cooker is sealed and placed on the heat. Given that the vapor pressure of water is $1bar$, calculate the boiling point of water.

Solutions:

$$P_{sat} = \left(\frac{t}{100}\right)^4 \Rightarrow \frac{t}{100} = (P_{sat})^{\frac{1}{4}} \Rightarrow t = 100 \times (P_{sat})^{\frac{1}{4}} = 100^\circ C$$

Exercise 2:

A mass of water ($m = 4g$) is introduced into an initially empty container of volume $V = 10l$ and heated to a temperature of $t_1 = 80^\circ C$.

Given for water : $P_{sat}(80^\circ C) = 0.466bar$; $P_{sat}(100^\circ C) = 1bar$

1. Will the saturating vapor pressure of water be reached inside the enclosure? Deduce the mass of water remaining in the liquid state.
2. The container is heated to a temperature of $t_2 = 100^\circ C$. What is the nature of the new equilibrium state? What will the vapor pressure be inside the enclosure?